

Normal operator

Lemma. Let $T: V \rightarrow V$ be a linear operator on fin. dim inner product space V .

If λ is an eigenvalue of T , then $\bar{\lambda}$ is an eigenvalue of T^* .

Proof. Let v be an eigenvector of λ . Then, for any $x \in V$,

$$0 = \langle 0, x \rangle = \langle (T - \lambda I)v, x \rangle = \langle v, (T - \lambda I)^*(x) \rangle$$

That is, $v \in (\text{Im}(T^* - \bar{\lambda}I))^{\perp} = \ker(T^* - \bar{\lambda}I) \neq \{0\}$.

Schur's Lemma

Let $T: V \rightarrow V$ be a linear operator on fin. dim inner product space V over a field F .

If a characteristic polynomial $f(t)$ of T splits over F ,

then there exists an orthonormal basis γ for V s.t. $[T]_{\gamma}$ is upper-triangular matrix.

Def. Let $T: V \rightarrow V$ be a linear operator on inner product space V .

If $T^* \circ T = T \circ T^*$, then T is called Normal operator

Proposition, Let $T: V \rightarrow V$ be a Normal operator on inner product space V .

1) For any $x \in V$, $\|T(x)\| = \|T^*(x)\|$.

2) For any $c \in F$, $T - cI$ is Normal.

3) For any $\lambda \in F$ and $x \in V$, $T(x) = \lambda x \Leftrightarrow T^*(x) = \bar{\lambda} x$

4) For any distinct eigenvalues λ_1, λ_2 of T , $x \in E_{\lambda_1}$ and $y \in E_{\lambda_2} \Rightarrow \langle x, y \rangle = 0$.

Hermitian

Let V be an Inner product space.

Prop. Let $T, U: V \rightarrow V$ be Hermitian.

$T \circ U$ is Hermitian iff $T \circ U = U \circ T$

Proof. $T \circ U$ is Hermitian $\Leftrightarrow T \circ U = (T \circ U)^* = U^* \circ T^* = U \circ T$.

Prop. Let $T: V \rightarrow V$ be a linear operator, and $W \subseteq V$ be a T -invariant subspace.

1. If T is Hermitian then a restriction $T|_W$ is Hermitian.

2. $W^\perp$ is T^* -invariant subspace.

3. If W is T, T^* -invariant, $T|_W^* = T^*|_W$.

4. If W is T, T^* -invariant and T is Normal, then $T|_W$ is Normal.

Proof. 1) is clear: for any $x, y \in W$,

$$\langle T|_W(x), y \rangle = \langle T(x), y \rangle = \langle x, T^*(y) \rangle = \langle x, T(y) \rangle = \langle x, T|_W(y) \rangle$$

2). Let $x \in W^\perp$ be given. Then, for $y \in W$,

$$\langle T^*(x), y \rangle = \langle x, T(y) \rangle = 0, \text{ because } T(y) \in W. \text{ Thus } T^*(x) \in W^\perp.$$

3). Let $x, y \in W$ be given. Then,

$$\begin{aligned} \langle T(x), y \rangle &\stackrel{1)}{=} \langle T|_W(x), y \rangle = \langle x, T|_W^*(y) \rangle \\ &\stackrel{2)}{=} \langle x, T^*(y) \rangle = \langle x, T^*|_W(y) \rangle. \end{aligned}$$

That is, for any $x, y \in W$,

$$\langle x, T|_W^*(y) - T^*|_W(y) \rangle = 0.$$

This implies $T|_W^* = T^*|_W$

4) Directly, given by 3.

Prop. Let V be a fin. dim. Complex inn prod. space, let $T: V \rightarrow V$ be a Normal.

If $W \subseteq V$ is T -invariant, then also T^* -invariant.

Proof. Since T is Normal operator on fin. dim. Comp. inn space, T is diagonalizable by an orthonormal β .

Let $x \in W$ be given with a representation $\sum a_i v_i$ where $v_i \in \beta$. Then,

$$T^*(x) = \sum a_i T^*(v_i) = \sum a_i \bar{\lambda}_i v_i \in W, \text{ since } v_i \in W.$$

Prop. If $T: V \rightarrow V$ is Normal, then

$$T(x) = 0 \Leftrightarrow \|T(x)\| = 0 \Leftrightarrow \|T^*(x)\| = 0 \Leftrightarrow T^*(x) = 0.$$

Thus, $\ker T = \ker T^*$.

Moreover, if V is fin. dim, then

$$\operatorname{Im} T = \operatorname{Im} T^*$$

Thm. Let V be a fin. dim, inn. space, $T: V \rightarrow V$ be a Hermitian.

For any $x \in V$,

$$\|T(x) \pm ix\|^2 = \|T(x)\|^2 + \|x\|^2.$$

Therefore, $T - iI$ is Invertible and $((T - iI)^{-1})^* = (T + iI)^{-1}$.

Proof. By lemma

$$\|T(x) \pm ix\|^2 = \|T(x)\|^2 + \|x\|^2 \pm 2\operatorname{Re}\langle T(x), ix \rangle$$

Meanwhile, for any $x \in V$,

$$\langle T(x), ix \rangle = -i \langle T(x), x \rangle$$

$$\langle T^*(ix), x \rangle = \langle x, iT(x) \rangle = -i \langle x, T(x) \rangle$$

This implies $\operatorname{Re} \langle T(x), ix \rangle = 0$.

Now,

$$\begin{aligned} \ker(T \pm iI) &= \{x \in V \mid (T \pm iI)(x) = 0\} \\ &= \{x \in V \mid \|T(x)\|^2 + \|x\|^2 = 0\} \\ &= \{0\}. \end{aligned}$$

Prop. Let V be a Complex inn. prod space.

1) T is Hermitian $\Leftrightarrow$ for all $x \in V$, $\langle T(x), x \rangle \in \mathbb{R}$.

2). If for all $x \in V$, $\langle T(x), x \rangle = 0$, then $T = T_0$.

Proof. At first, show that 2). For any $x, y \in V$,

$$0 = \langle T(x+y), x+y \rangle = \langle T(x), x \rangle + \langle T(x), y \rangle + \langle T(y), x \rangle + \langle T(y), y \rangle = \langle T(x), y \rangle + \langle T(y), x \rangle$$

$$0 = \langle T(x+iy), x+iy \rangle = \langle T(x), x \rangle + \langle T(x), iy \rangle + \langle T(iy), x \rangle + \langle T(iy), iy \rangle = \langle T(x), iy \rangle + \langle T(iy), x \rangle$$

$$\text{Now, } \langle T(x), y \rangle + \langle T(y), x \rangle = 0 \text{ and } -\langle T(x), y \rangle + \langle T(y), x \rangle = 0,$$

That is, $\langle T(x), y \rangle = 0$. Since x, y are chosen arbitrarily, $T = T_0$.

Suppose T is Hermitian. Then, for any $x \in V$,

$$\langle T(x), x \rangle = \langle x, T^*(x) \rangle = \langle x, T(x) \rangle = \overline{\langle T(x), x \rangle}$$

Thus $\langle T(x), x \rangle \in \mathbb{R}$.

Conversely, suppose for all $x \in V$, $\langle T(x), x \rangle \in \mathbb{R}$. Then, directly $\langle T^*(x), x \rangle \in \mathbb{R}$.

$$\begin{aligned} \text{Now } \langle (T - T^*)(x), x \rangle &= \langle T(x), x \rangle - \langle T^*(x), x \rangle \\ &= \langle T(x), x \rangle - \langle x, T(x) \rangle \\ &= 0. \end{aligned}$$

By 2), we obtain the result.

Prop. $A = B^t B$ for some $B \in M_n(\mathbb{R})$, A is called Gramian matrix.

A is Gramian if and only if $A^t = A$ and A has no negative eigenvalues.

Proof. Suppose that A is Gramian. Then, clearly symmetric.

$$\text{Since } L_A = L_{B^t B} = (L_{B^t}) \circ (L_B),$$

$$\langle L_A v, v \rangle = \langle L_{B^t B} v, v \rangle \geq 0.$$

Conversely, For each $1 \leq i, j \leq n$,

$$([U^*]_{\beta})_{i,j} = \langle U^*(v_i), v_j \rangle = \langle v_i, U(v_j) \rangle = \sqrt{\lambda_i} \cdot \langle v_i, v_j \rangle = \sqrt{\lambda_i} \delta_{i,j}$$

$$\text{Meanwhile, } T(v_i) = (U^* \circ U)(v_i) \quad [T]_{\beta} = A = [U]_{\beta}^* [U]_{\beta}.$$

Def. Let V be a fin. dim. inn. space, $T: V \rightarrow V$ be a Hermitian.

T is called positive semidefinite if

$$\forall x \in V, \langle T(x), x \rangle \geq 0$$

In addition, if $\langle T(x), x \rangle = 0 \Leftrightarrow x = 0$, then T is called positive definite.

Thm. Let V be an inn. space with $\dim V = n$, and β be an orthonormal basis for V .

Let $T: V \rightarrow V$ be Hermitian, $A = [T]_{\beta}$.

1). T is positive semidefinite if and only if every eigenvalue of T is non-negative.

2) T is positive definite if and only if every eigenvalue of T is positive.

Proof. Let γ be an orthonormal basis consisted by eigenvectors of T .

Let $x \in V$ be given. Let $\gamma = \{v_1, \dots, v_n\}$, $B = [T]_{\gamma}$. Then,

$$x = \sum_{i=1}^n a_i v_i \quad \text{for some } a_i \in F.$$

$$\begin{aligned} \text{Now, } \langle T(x), x \rangle &= \langle T(\sum_{i=1}^n a_i v_i), \sum_{i=1}^n a_i v_i \rangle \\ &= \langle \sum_{i=1}^n a_i \lambda_i v_i, \sum_{i=1}^n a_i v_i \rangle \\ &= \lambda_i \cdot \langle \sum_{i=1}^n a_i v_i, \sum_{i=1}^n a_i v_i \rangle \\ &= \sum_{i=1}^n \lambda_i \cdot \|a_i v_i\|^2 \end{aligned}$$

3). T is positive definite if and only if for any $(a_1, \dots, a_n) \in F^n$, $\sum_{i=1}^n A_{ii} a_i \bar{a}_i > 0$.

Proof. Put $\beta = \{w_1, \dots, w_n\}$. Then, for $\underline{\sum a_i w_i}$,

$$\begin{aligned} &\langle T(\sum a_i w_i), \sum a_i w_i \rangle \\ &= \sum_{i,j} a_i \bar{a}_j \langle T(w_i), w_j \rangle \\ &= \sum \underline{A_{ji}} a_i \bar{a}_j \end{aligned}$$

Prop. If $T^2 = U^2$ and T, U are positive semidefinite, then $T = U$.

Proof. Let $\beta = \{v_1, \dots, v_n\}$ be an orthonormal basis of $\mathbb{R}^n$.

For each $1 \leq i \leq n$, $T^2(v_i) = \lambda_i^2 v_i = U^2(v_i)$.

If $\lambda_i = 0$, then clearly $T(v_i) = 0$ and

$$0 = \langle U^2(v_i), v_i \rangle = \langle U(v_i), U(v_i) \rangle$$

Thus $U(v_i) = 0$.

Now, assume $\lambda_i > 0$.

$$0 = (U^2 - \lambda_i^2 I)(v_i) = (U + \lambda_i I)(U - \lambda_i I)(v_i)$$

Since U is positive semidefinite, every eigenvalue is bigger than zero.

Thus $U + \lambda_i I$ is invertible. Now, $(U - \lambda_i I)(v_i) = 0$. $\therefore U(v_i) = \lambda_i v_i = T(v_i)$.